



SHIRE HIGHLANDS EDUCATION DIVISION

2024 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

AGRICULTURE

PAPER I (100 marks)

PROVISIONAL MARKING SCHEME

SECTION A (70 Marks)

1. a. Removal of excess water from the soil. (1 mark)

b. Sub surface, surface, bio- drainage (2 marks for any two)

C .Mulching soil / planting vegetative cover/(1) planting trees which helps to reduce the speed of run – off controlling soil erosion (1)

Applying manure(1) which improve soil fertility(1)

Practicing crop rotation / including leguminous crops in rotation or including covers crops in rotation (1) this reduces speed of runn off controlling soil erosion/ fix nitrogen(1)

Practicing agroforestry (1) which helps to add soil fertility (1)

2. a (i) Anthracnose (1 mark)

(ii) Applying fungicides such as benlate (1 mark)

(iii) Source of foreign currency after exporting to other country(1) hence improving economic status of the country

source of employment in mango orchard or mango processing industry(1) hence improving living standard of people (1)

Source of income(1) after selling(1) (4 marks for any two)

3. (i)

$$PED = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

$$\% \text{ change in quantity supplied} = \frac{(\text{Original quantity}) - \text{new quantity} \times 100}{\text{original quantity}}$$

$$= \frac{(400-300)units \times 100}{400} \quad (1)$$

$$= \frac{100 \times 100}{400}$$

$$= 25\% \quad (1)$$

$$\% \text{ change in price} = \frac{\text{Original price} - \text{new price}) \times 100}{\text{original price}}$$

$$= \frac{(MK60 - MK80) \times 100}{MK60} \quad (1)$$

$$= 33.3\% \quad (1)$$

$$\text{EPD} = \begin{matrix} 25\% \\ 33.3\% \end{matrix}$$

$$0.75 \quad (1)$$

(ii) inelastic demand (1)

4. (i) animal power (1 mark)

(ii) They may damage the crops

They may be affected by weather conditions

They require extra cost in case of sickness

Animals may not copy up with large acreage of land compare to tractors

(1 mark for any one)

(iii)

- Farmers should not be harsh to animals/farmers should properly train animals
- Animals should not be exposed to midday heat. Early day or late day operation is recommendable.
- Carts should not be overloaded with farm produce.
- Animals should be given enough and proper diet

(2 marks for any two)

(iv) easy transportation of farm produce and inputs **(1 mark)**

(v) State any two ways of maintaining the farm implement.

- Lubricate all moving parts using oil or grease to reduce friction
- Replace worn out parts
- Repair broken parts

- Tighten loose bolts and nuts
- Clean the implements after use
- Store equipment at dry place
- Make sure that tyres are at the correct pressure. (1 mark for any one)

5. a. Improving uniformity/colour/size/shape

Increasing nutritive value of crops

Improving processing qualities

(2 marks for any two)

b. (i) Introduction (1 mark)

(ii) Differences in environmental conditions. (1 mark)

6. (a) It is a cheaper source of feed(1) compared to commercial feed(1)

- Pasture help to control soil erosion since the roots bind soil particles(1)
- Leguminous pasture help to maintain soil fertility(1) since they fix nitrogen into the soil
- Pasture help to control pests and diseases (1) sine they break their life cycle(1)
- Pasture help to maintain soil structure when the leaves and roots decay to form organic matter. (1) (4 marks)

b. They contain high dry matter content hence high yield per hector

- They contain high dry matter content hence high yield per hector
- They have high protein content which improves nutrition status of livestock
- They are typically more digestible and palatable which improves appetite of livestock (4 marks)

c. (i)It helps to increase soil PH legumes plants perform better in medium basic soil and perform poorly in acidic soil.

Liming improve survival rate of rhizobium bacteria hence providing effective nodulation. (1 mark)

(ii) Hulling increase the rate of seed germination

Makes pelleting of seed easier. (1 mark)

7. a. Seed rate= *expected plant per ha* ÷ *seedarte* ÷ *purity%* ÷ *germination%*

$$= \frac{800000 \text{ plants} \times 1\text{kg} \times 100 \times 100}{\text{ha} \times 400000 \text{ seeds} \times 40 \times 80} \quad (1)$$

= 6.25kg/ha

(1)

b. (i) Seed rate will increase

(1 mark)

(ii) To achieve the desired plant population(1) since the seed is mixed with trash(1)

8. a. Tick **(1 mark)**

b.

- Suck blood (1)and causing anaemia.(1)
- Damage hides through their biting wounds(1) and reduces quality of hides(1)
- The bite wounds may become a source of infection/ mastitis(1) this reduces production(1)
- Ticks transmit four tick borne diseases/East Coast Fever (ECF)/ Heart water/ red water and gall sickness(1) which can lead to death of cattle.(1)

(4 marks for any two)

c. burning infested pasture

practicing rotation grazing (**1 mark for any one**)

9. a. The ability of the soil to hold and release cations (**1 mark**)

b. Applying manure /Applying organic manure/applying lime (**2 marks**)

c. The high the bulk density, the low the porosity of the soil. (**1 mark**)

d. (i) A gihly consistency soil does not promote root penetration and development(1) hence low maize production(1)

A highly consistency soil is poorly aerated(1) hence poor root and microbial respiration(1)

A highly consistency soil becomes waterlogged(1) hence causes denitrification and maize lack nitrogen.(1)

(4 marks for any two)

10.

a.(i) Cap/ pileus(**1 mark**)

(ii) For production of spawn/seed. (**1 mark**)

b. Maize stalks, rice straws, banana leaves, cotton wastes, sawdusts, maize cobs (**2 marks for any two**)

13. Farmers in Traditional authority Kalima grow maize every year and they are privileged with the following resources:

- Soil testing unit.
- Credit cooperatives (SACCO).
- Agriculture Extension office nearby
- A big river passing through the village

Despite of having the above resources they face hunger every year Explain **any five ways** in which farmers can utilize resources to ensure food security.

- Soil testing unit would help farmer to know soil ph of their farms in order to select suitable crops to be grown so that they produce more food/.Soil testing unit helps the farmer to know texture type so that they can select suitable soil for their crops so that they produce more food.
- Farmers borrow money from credit cooperatives and use them to buy farm inputs/ Improved seeds/ fertilizer(1) in order to produce more food(1)
- Credit cooperatives allow farmers to keep their after selling their farm produce for future use
- Farmers seek knowledge from extension office on how they can plan and budget their farming operations.
- Extension officers train farmers on various new farming practices(1) so that they produce more food(1)
- The river helps the farmer to cultivate more than twice a year(1) through irrigation(1)